Municipal UAS Platform — Pricing & Calculation Model

Formula Framework (Structure, Not Numbers)

Purpose: Define the *underlying formulas* used to price a shared municipal drone platform and to quantify its value. All quantities are expressed as variables so the model can be populated with real figures during a costing exercise. No indicative prices are stated.

Table of Contents

1. Modeling Principles

- **Shared asset, allocated cost.** The platform is one cost pool; departments are charged in proportion to their use (Section 5).
- **Cost floor, value ceiling.** Pricing must recover Total Cost of Ownership (TCO) and a margin; the *justification* to the client is value/cost-avoidance (Section 7).
- **Per-unit transparency.** Everything resolves to a defensible **cost per flight hour** and **cost per mission**, which procurement can audit.
- **Time-phased.** Capital is amortized; operating costs recur. All multi-year figures are discounted to present value.

2. Notation & Variable Glossary

Symbol	Meaning	Unit
`C_cap`	Total capital expenditure (airframes, payloads, ground stations)	currency
`C_soft`	Software & licensing cost (annual)	currency/yr
`C_pilot`	Crew/labour cost (annual, loaded)	currency/yr
`C_train`	Training & certification cost (annual)	currency/yr
`C_maint`	Maintenance, spares & repairs (annual)	currency/yr

Symbol	Meaning	Unit
C_{insure}	Insurance & liability cover (annual)	currency/yr
C_{reg}	Regulatory, compliance & airspace fees (annual)	currency/yr
C_{data}	Data storage, processing & connectivity (annual)	currency/yr
C_{gov}	Governance, audit & privacy program (annual)	currency/yr
C_{over}	General overhead & admin (annual)	currency/yr
N	Asset useful life	years
r	Discount rate	fraction/yr
R	Residual / salvage value at end of life	currency
H	Total available flight hours per year (capacity)	hours/yr
U	Utilization rate (billable ÷ available)	fraction
H_b	Billable flight hours per year = $H \cdot U$	hours/yr
m	Margin / contribution rate	fraction
d	Department index	—
w_d	Department d's share of usage	fraction
V	Quantified value / benefit	currency

3. Cost Model (TCO)

3.1 Amortized capital (annualized capex)

Using a capital recovery (annuity) factor so capex is spread over the asset life:

$$CRF = [\, r \cdot (1 + r)^N \,] \,/\, [\, (1 + r)^N - 1 \,]$$

$$C_{cap\_annual} = (C_{cap} - R \,/\, (1 + r)^N) \cdot CRF$$

3.2 Annual operating cost

$$C_{op} = C_{soft} + C_{pilot} + C_{train} + C_{maint} + C_{insure} + C_{reg} + C_{data} + C_{gov} + C_{over}$$

3.3 Total annual cost of ownership

$$TCO_{annual} = C_{cap\_annual} + C_{op}$$

3.4 Multi-year present value (optional)

$$TCO_{PV} = \Sigma\ (t = 1 \ldots N)\quad TCO_{annual}(t) \,/\, (1 + r)^t$$

4. Capacity & Utilization Model

4.1 Available flight hours per year

$$H = D_{op} \cdot F \cdot h_f \cdot A$$

Where:

- D_{op} = operational days per year
- F = flights per operational day

- h_f = average billable hours per flight
- A = fleet availability factor (uptime after maintenance/weather), $0 < A \le 1$

4.2 Billable hours

$$H_b = H \cdot U$$

4.3 Unit cost per flight hour (the keystone metric)

$$Cost\_per\_hour = TCO\_annual / H_b$$

> *Note the lever: as utilization* U *rises (more departments sharing the asset),* `Cost_per_hour` *falls. This is the core economic argument of the shared platform.*

4.4 Cost per mission

$$Cost\_per\_mission = Cost\_per\_hour \cdot h\_mission + C\_consumables + C\_payload\_specific$$

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5. Cost Allocation Across Departments

Each department d carries a usage weight w_d (e.g., share of billable hours), with $\Sigma\ w_d = 1$.

5.1 Usage weight

$$w_d = H_{b,d} / H_b \qquad \text{(where } H_{b,d} = \text{billable hours consumed by department } d\text{)}$$

5.2 Allocated cost to a department

$$C_d = TCO\_annual \cdot w_d$$

5.3 Optional two-part allocation (fixed base + variable use)

To reflect that some cost is fixed regardless of use:

$$C_d = (TCO\_fixed / n\_departments) + (TCO\_variable \cdot w_d)$$

Where `TCO_fixed` covers governance, base licensing, and minimum crew, and `TCO_variable` scales with flight hours (fuel/battery, wear, mission-specific data processing).

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6. Pricing Models

Pricing = cost recovery + margin. Choose the model (or blend) that fits the client's procurement preference.

6.1 Cost-plus per flight hour

$$Price\_per\_hour = Cost\_per\_hour \cdot (1 + m)$$

6.2 Per-mission / fixed-fee

$$Price\_mission = Cost\_per\_mission \cdot (1 + m)$$

6.3 Subscription / availability fee (Drone-as-a-Service)

A recurring fee guaranteeing capacity, plus a usage charge above an included allowance:

$$Price\_subscription\_annual = TCO\_annual \cdot (1 + m)$$

$$Overage\_charge = \max(0, (H\_used - H\_included)) \cdot Price\_per\_hour$$

6.4 Tiered subscription

$$Price\_tier(k) = Base\_k + Per\_hour\_k \cdot \max(0, H\_used - Included\_k)$$

Where tier k bundles an included-hours allowance $Included_k$ with a base fee $Base_k$ and a reduced marginal Per_hour_k.

6.5 Outcome / SLA-linked component (optional)

Adjust price by performance against a service-level target (e.g., response-time or availability KPI):

$$Price\_effective = Price\_base \cdot (1 + \alpha \cdot (KPI\_actual - KPI\_target) / KPI\_target)$$

Where α is the agreed sensitivity (bonus/penalty band), bounded by a cap and floor.

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7. Value, Cost-Avoidance & ROI Model

This is the **justification** layer presented to the client — it does not set price but proves worth.

7.1 Cost avoidance vs. legacy methods

$$V\_avoid = \Sigma\ (legacy\_method\_cost - uas\_method\_cost)$$

Examples of legacy methods replaced: manned-helicopter hours, scaffolding/cherry-picker inspections, road-closure costs, external survey contractors.

7.2 Time-savings value

$$V\_time = \Sigma\ (\Delta t_i \cdot rate_i)$$

Where Δt_i is time saved on activity i (e.g., faster road clearance, faster inspection) and $rate_i$ is the value of that time (labour, congestion cost, downtime cost).

7.3 Risk / safety value (expected-loss reduction)

$$V\_safety = \Sigma\ (P\_before - P\_after) \cdot L$$

Where P is incident probability and L is the expected loss per incident (harm to personnel or public avoided by keeping crews out of hazard).

7.4 Recovery & revenue value

$$V\_recovery = improved\_claim\_value + reduced\_downtime\_value + enforcement\_recovery$$

7.5 Total quantified value

$$V\_total = V\_avoid + V\_time + V\_safety + V\_recovery$$

7.6 Return on Investment

$$ROI = (V\_total - TCO\_annual) / TCO\_annual$$

7.7 Payback period

$$Payback = C\_cap / (V\_total\_annual - C\_op)$$

7.8 Benefit–cost ratio

$$BCR = V\_total\_PV / TCO\_PV$$

Where both numerator and denominator are discounted to present value over the asset life.

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8. Risk, Contingency & Sensitivity

8.1 Contingency-loaded cost

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TCO_loaded = TCO_annual · (1 + c)
```

Where `c` is a contingency rate for weather downtime, attrition, and regulatory change.

8.2 Break-even utilization

The utilization at which price exactly recovers cost:

```
U_breakeven = TCO_annual / (H · Price_per_hour / (1 + m))
```

8.3 Sensitivity

Test how `Cost_per_hour`, `ROI`, and `Payback` move when each driver flexes ±x%: `U`, `A`, `H`, `C_cap`, `C_pilot`, and `r`. Report the variables to which the model is most sensitive (typically `U` and `C_pilot`).

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9. Worked Formula Walkthrough

Order of computation (plug real inputs in this sequence):

1. `CRF` → annualize capital → `C_cap_annual` *(3.1)*
2. Sum operating lines → `C_op` *(3.2)*
3. `TCO_annual = C_cap_annual + C_op` *(3.3)*
4. Build capacity `H`, apply utilization → `H_b` *(4.1–4.2)*
5. `Cost_per_hour = TCO_annual / H_b` *(4.3)*
6. Apply margin → `Price_per_hour` and/or subscription *(6)*
7. Allocate to departments via `w_d` *(5)*
8. Quantify `V_total`, then `ROI`, `Payback`, `BCR` *(7)*
9. Stress-test with contingency, break-even, sensitivity *(8)*

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10. Inputs Checklist

Collect these before populating the model:

- [ ] Capital line items (airframes, payloads, ground control) → `C_cap`, `R`
- [ ] All annual operating line items → `C_soft … C_over`
- [ ] Asset life `N`, discount rate `r`, contingency `c`
- [ ] Operational profile: `D_op`, `F`, `h_f`, availability `A`
- [ ] Expected utilization `U` and per-department usage weights `w_d`
- [ ] Target margin `m` and pricing model choice (6.1–6.5)
- [ ] Legacy-method baseline costs (for `V_avoid`)
- [ ] Time, safety, and recovery parameters (for `V_time`, `V_safety`, `V_recovery`)
- [ ] KPI targets if using SLA-linked pricing (6.5)

Companion document: "Municipal UAS Platform — Capability & Use-Case Catalogue."